Safe Navigation using Hamilton-Jacobi Reachability Analysis with Control Lyapunov-Value Functions

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INTRODUCTION

Safety is of utmost importance in engineering and autonomous systems. Safety can be defined as the absence of undesirable events, while liveness represents the achievement of desirable outcomes. Especially for autonomous vehicles that interact with dynamic urban environments, ensuring consistent reliability and dependability is imperative. In the past, control theory for such systems primarily emphasized ensuring liveness. However, the significance of safety, particularly in intricate autonomous systems, has grown considerably. This project addresses the need for a unified approach that combines Hamilton-Jacobi (HJ) reachability analysis and Control Lyapunov-Value Functions (CLVFs) to ensure the safe and live navigation of autonomous vehicles.

PROBLEM

Fig. 1: The planning system (dashed blue line) uses a simplified planning model, but the real vehicle cannot track this path exactly (solid red line). Figure obtained from [1]

Previous path planning methods for dynamical systems that utilize HJ reachability and Control Lyapunov Functions (CLFs) suffer from computational complexity. To enable real-time planning, several algorithms opt for significantly simplified model dynamics or kinematics [1]. This has led to a tracking error between the intended path and the actual high-dimensional system as shown in Fig. 1.

PROPOSED SOLUTION

We seek to demonstrate the practical implementation of an integrated approach of CLFs using HJ reachability in autonomous vehicle systems. We can do this by constructing CLVFs to generate safe navigation of a Dubins car.

HAMILTON-JACOBI REACHABILITY ANALYSIS

HJ Reachability Analysis determines the set of states from which a system can reach a target set under given constraints, using the Hamilton-Jacobi-Bellman PDE to compute the value function that defines the reachable set. The focus is on calculating the reach-avoid set (Fig. 3), which identifies states from which the system can reach a target set while avoiding unsafe regions over a specified time [2].

Consider the following 3D Dubins car:

$$\dot{x}_1 = v\cos(x_3), \dot{x}_2 = v\sin(x_3), \dot{x}_3 = u$$

where $v = 1$ is the constant velocity, $x = [x_1, x_2, x_3]^T$ are the states denoting x-direction, y-direction and heading angle, and $u \in [-1,1]$ is the control. Consider the target set $\mathcal{T}$, which is a radius 2 circle in the $x_1 - x_2$ plane, centered at the origin. The goal is to stay within $\mathcal{T}$ for 5 seconds.

The value function can be defined as:

$$V(x,T) = \min_{u} \max_{t\in[0,T]} \ell(\mathcal{E}(t;0,x,u))$$

$$V(x,5) = \min_{u} \max_{t\in[0,5]} \ell(\mathcal{E}(t;0,x,u))$$

Using MATLAB toolboxes ToolboxLS and helperOC, the evolution of the value function can be simulated.

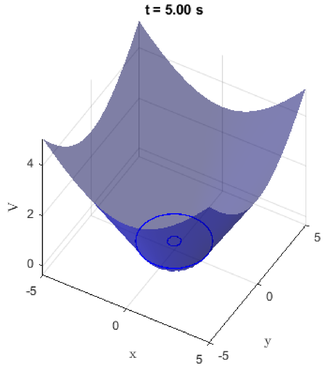


Fig. 2: HJ Value Function Visualization

The HJPDE is:

$$0 = \max\{\ell(x) - V(x,T), \frac{dV}{dt} + \min_{u} \frac{dV}{dt} \cdot f(x,u)\}$$

$$V(x,5) = \ell(x)$$

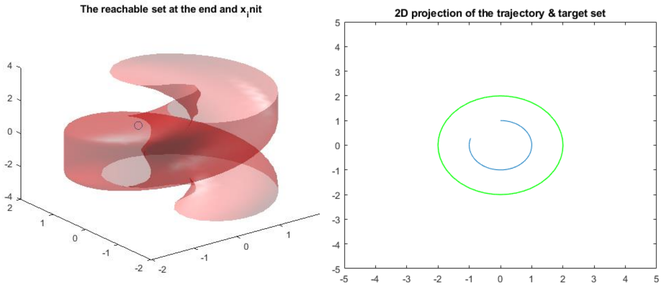


Fig. 3: Reachable set (left) and 2D plot showing the optimal trajectory of the Dubins car projected onto the $x_1 - x_2$ plane (blue arc), along with the target set (green circle) on the right.

CONTROL LYAPUNOV VALUE FUNCTIONS

The CLVF discovers and stabilizes the smallest control invariant set around a specified point. It is the viscosity solution to a modified HJ variational inequality [3] and identifies the region of exponential stability based on input constraints and an exponential decay rate γ.

Consider the same Dubins car with $u \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and the loss function $\ell(x) = \|x\|_2$. The control will minimize the loss.

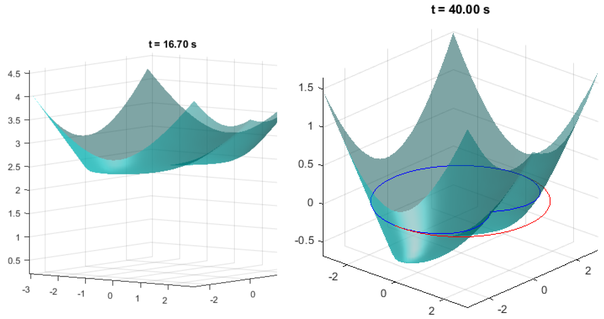


Fig. 5: CLVF Value Function Visualization with $\gamma = 0$ (left) and $\gamma = 0$ (right)

When $\gamma = 0$, the solution converges to steady state at $t = 16.70s$ according to user-defined convergence threshold. When $\gamma = 1$, the solver iterates for the full time horizon.

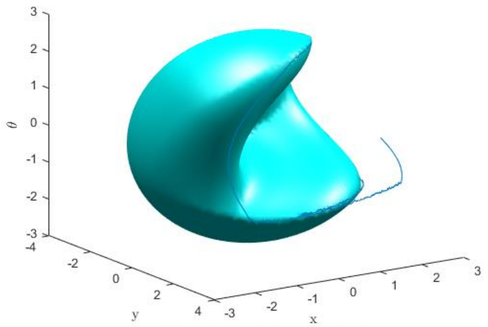


Fig. 6: Reachable Set and Trajectory

The reachable set (cyan shape) visualizes how the set of safe states evolves over time. The trajectory (dark blue line) indicates the path taken by the car under the optimal control strategy.

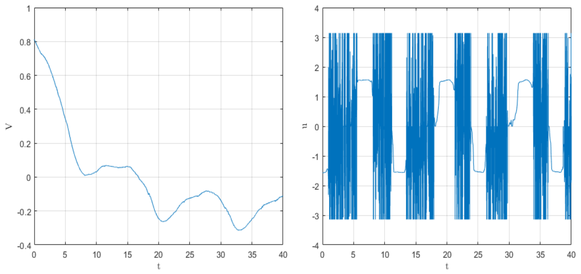


Fig. 7: 2D Subplots of V and $u(t)$

The value function V evolves over time (left). A decreasing value function suggests that the system is reaching the target set. The control input $u(t)$ is applied to the car to minimize V and ensure it follows the desired trajectory.

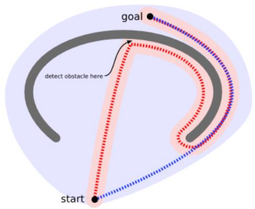
CONCLUSIONS

Guaranteeing safety in autonomous vehicles is crucial for their broader adoption and integration into everyday transportation. This project utilized Hamilton-Jacobi reachability analysis and Control Lyapunov-Value Functions to develop a unified approach for safe and live navigation of autonomous vehicles. Our findings demonstrate that this integrated method significantly reduces tracking errors and enhances real-time path planning accuracy for high-dimensional systems. Our future research will focus on implementing these algorithms into hardware to validate their effectiveness in real-world scenarios.

EXTENDED RESEARCH

The practical implementation of HJ Reachability and CLVFs may prove useful to existing path planning methods such as the A* path planning algorithm.

A* is a widely used pathfinding and graph transversal algorithm. It is employed to find the shortest path from a starting point to a goal point on a graph or grid [4]. It can serve to efficiently plan an obstacle-free path.



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[1] S. Herbert, M. Chen, S. Han, S. Bansal, J. Fisac, and C. Tomlin, "FaSTrack: a Modular Framework for Fast and Guaranteed Safe Motion Planning," Feb 2021.

[2] S. Bansal, M. Chen, S. Herbert, and C. Tomlin, "Hamilton-Jacobi Reachability: A Brief Overview and Recent Advances," IEEE Conference on Decision and Control, Sept 2017.

[3] S. Bansal, M. Chen, S. Herbert, and C. Tomlin, "Hamilton-Jacobi Reachability: A Brief Overview and Recent Advances," IEEE Conference on Decision and Control, Sept 2017.

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